

HAWARDEN, United KingdomWMO: **033210**Lat: **53.175N**Lon: **2.987W**Elev: **14**StdP: **101.16**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.5	-5.9	2.3	-1.1	-4.5	2.6	0.0	13.3	11.1	11.8	10.2	1.0	160	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	25.3	18.2	23.2	17.4	21.6	16.6	19.2	23.3	18.2	22.0	17.3	20.6	3.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.7	20.9	16.7	11.9	19.9	15.8	11.3	19.0	54.5	23.1	51.4	22.1	48.6	20.7	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	9.0	7.9	-7.4	28.5	3.0	2.1	-9.5	30.0	-11.2	31.3	-12.9	32.4	-15.1	33.9		
			-7.5	22.5	2.2	2.4	-9.0	24.2	-10.3	25.6	-11.5	26.9	-13.1	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	5.0	5.4	6.7	8.7	11.6	14.6	16.4	16.2	13.9	10.9	7.6	5.4
	DBStd	5.03	3.44	3.23	2.96	2.82	2.76	2.43	2.29	2.15	2.23	2.90	3.12	3.82
	HDD10.0	721	156	131	107	58	14	1	0	0	1	23	83	148
	HDD18.3	2979	413	362	359	290	208	116	71	74	133	229	322	402
	CDD10.0	804	2	3	6	18	64	137	197	191	118	53	11	4
	CDD18.3	20	0	0	0	0	0	3	9	7	1	0	0	0
	CDH23.3	173	0	0	0	0	7	33	81	45	6	1	0	0
CDH26.7	27	0	0	0	0	0	4	13	9	0	0	0	0	
Wind	WSAvg	3.6	4.1	4.2	4.0	3.6	3.5	3.4	3.2	3.2	3.2	3.4	3.6	3.8
Precipitation	PrecAvg	703	60	49	55	45	56	53	46	56	62	68	80	73
	PrecMax	862	96	121	93	89	159	112	110	101	257	140	185	141
	PrecMin	510	17	2	8	2	11	14	12	3	5	11	38	24
	PrecStd	79	21	28	23	25	37	25	25	24	49	31	35	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	14.4	16.2	20.9	24.3	27.2	28.5	27.9	24.1	19.5	16.0	15.2
		MCWB	11.5	11.7	11.2	13.6	16.8	18.8	19.1	19.0	17.8	15.9	13.8	12.2
	2%	DB	12.5	12.7	14.2	17.4	21.3	23.5	25.5	23.9	21.5	17.8	14.7	13.3
		MCWB	10.4	10.4	10.5	12.2	15.4	17.1	18.4	18.0	16.5	14.9	12.7	11.3
	5%	DB	11.3	11.6	12.8	15.4	18.7	21.3	23.3	21.9	19.7	16.6	13.5	12.2
		MCWB	9.4	9.4	9.8	11.2	13.9	16.1	17.7	17.3	15.5	14.1	11.7	10.4
	10%	DB	10.2	10.5	11.7	13.8	16.9	19.6	21.3	20.4	18.3	15.6	12.5	10.9
		MCWB	8.4	8.4	9.0	10.3	12.8	15.1	16.7	16.3	14.6	13.3	10.7	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	12.3	12.2	14.4	17.9	20.6	21.6	20.4	18.8	16.8	14.3	12.8
		MCDB	13.4	13.8	14.8	18.8	22.9	23.9	25.2	24.4	22.6	18.8	15.6	14.0
	2%	WB	10.7	10.7	11.1	13.0	16.0	18.1	19.2	18.8	17.4	15.6	13.1	11.7
		MCDB	12.3	12.4	13.5	16.1	20.4	21.8	23.6	22.6	20.6	17.2	14.4	13.2
	5%	WB	9.7	9.7	10.2	11.8	14.6	16.7	18.1	17.9	16.2	14.6	12.0	10.7
		MCDB	11.1	11.4	12.5	14.7	17.9	20.5	22.2	21.3	18.8	16.2	13.3	12.1
	10%	WB	8.6	8.6	9.3	10.7	13.4	15.6	17.1	17.1	15.2	13.6	10.9	9.3
		MCDB	10.0	10.3	11.4	13.4	16.2	18.9	20.8	19.7	17.6	15.2	12.2	10.7
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.2	7.4	8.7	8.9	8.5	8.6	8.2	8.3	7.2	6.1	5.7
		MCDBR	6.5	6.8	9.0	11.4	12.5	12.4	13.0	10.9	10.6	7.9	6.5	6.4
		MCWBR	5.7	5.4	6.2	7.1	7.5	7.4	7.1	5.9	6.4	5.6	5.5	5.6
	5% WB	MCDBR	6.3	6.6	8.4	10.2	11.5	10.7	11.2	9.6	9.2	7.5	6.2	6.1
		MCWBR	5.8	5.5	6.3	6.9	7.4	7.1	7.0	5.8	6.1	5.9	5.5	5.6
Clear-Sky Solar Irradiance	taub	0.326	0.341	0.372	0.396	0.400	0.396	0.396	0.381	0.369	0.351	0.326	0.316	
	taud	2.318	2.367	2.288	2.239	2.271	2.313	2.320	2.377	2.410	2.423	2.379	2.292	
	Ebn at Noon	639	758	805	832	848	855	847	841	805	737	639	571	
	Edh at Noon	61	79	103	123	125	121	119	106	91	73	57	52	
All-Sky Solar Radiation	RadAvg	0.66	1.33	2.48	3.83	4.84	5.09	4.80	3.93	2.85	1.59	0.80	0.50	
	RadStd	0.06	0.14	0.23	0.37	0.33	0.49	0.49	0.27	0.21	0.14	0.08	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page